

Exercise 4: Launching Web Applications using App Engine Tools

Aim:

To launch and run a web application locally using Google App Engine tools (Flask) and verify it in a browser.

Procedure:

Step 1: Upgrade Flask and Werkzeug

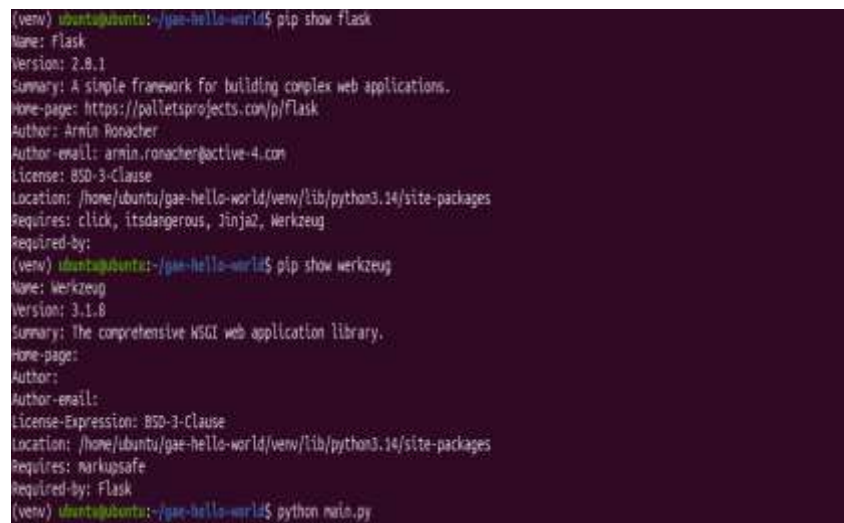
```
pip install --upgrade flask werkzeug
```

This upgrades Flask and its dependencies (Werkzeug, blinker, etc.) in the virtual environment.

Step 2: Verify Installed Package Versions

```
pip show flask
```

```
pip show werkzeug
```

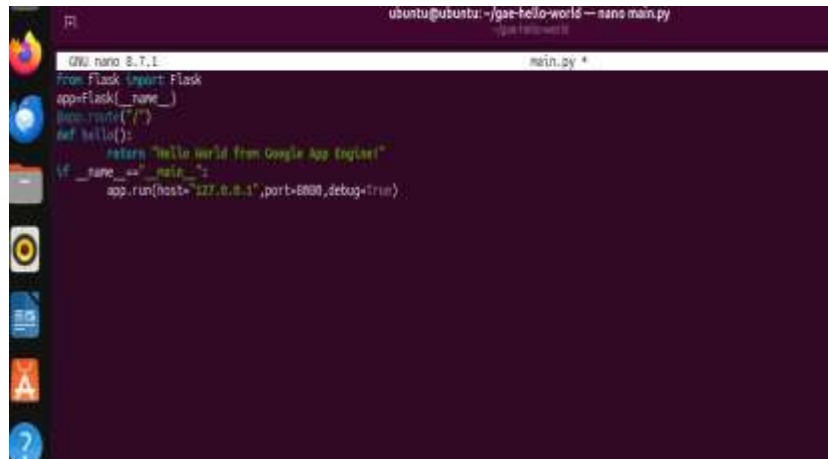


```
(venv) ubuntu@ubuntu:~/gae-hello-world$ pip show flask
Name: flask
Version: 2.0.1
Summary: A simple framework for building complex web applications.
Home-page: https://palletsprojects.com/p/flask
Author: Armin Ronacher
Author-email: armin.ronacher@active4.com
License: BSD-3-Clause
Location: /home/ubuntu/gae-hello-world/venv/lib/python3.14/site-packages
Requires: click, itsdangerous, Jinja2, Werkzeug
Required-by:
(venv) ubuntu@ubuntu:~/gae-hello-world$ pip show werkzeug
Name: Werkzeug
Version: 3.1.8
Summary: The comprehensive WSGI web application library.
Home-page:
Author:
Author-email:
License-Expression: BSD-3-Clause
Location: /home/ubuntu/gae-hello-world/venv/lib/python3.14/site-packages
Requires: MarkupSafe
Required-by: Flask
(venv) ubuntu@ubuntu:~/gae-hello-world$ python main.py
```

Step 3: Review Application Code (main.py)

main.py defines a simple Flask route that returns a Hello World message:

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Harish Kumar V

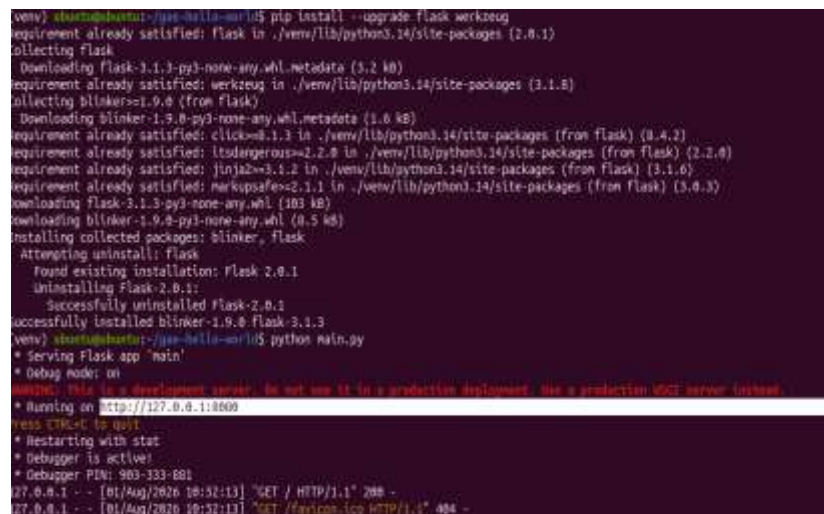


```
ubuntu@ubuntu: ~/gse-hello-world -- nano main.py
GNU nano 2.7.1 main.py
from flask import Flask
app=Flask(__name__)
@app.route("/")
def hello():
    return "Hello world from Google App Engine!"
if __name__=="__main__":
    app.run(host="127.0.0.1",port=8080,debug=True)
```

Step 4: Run the Application

python main.py

Flask starts the development server on <http://127.0.0.1:8080> with debug mode enabled.



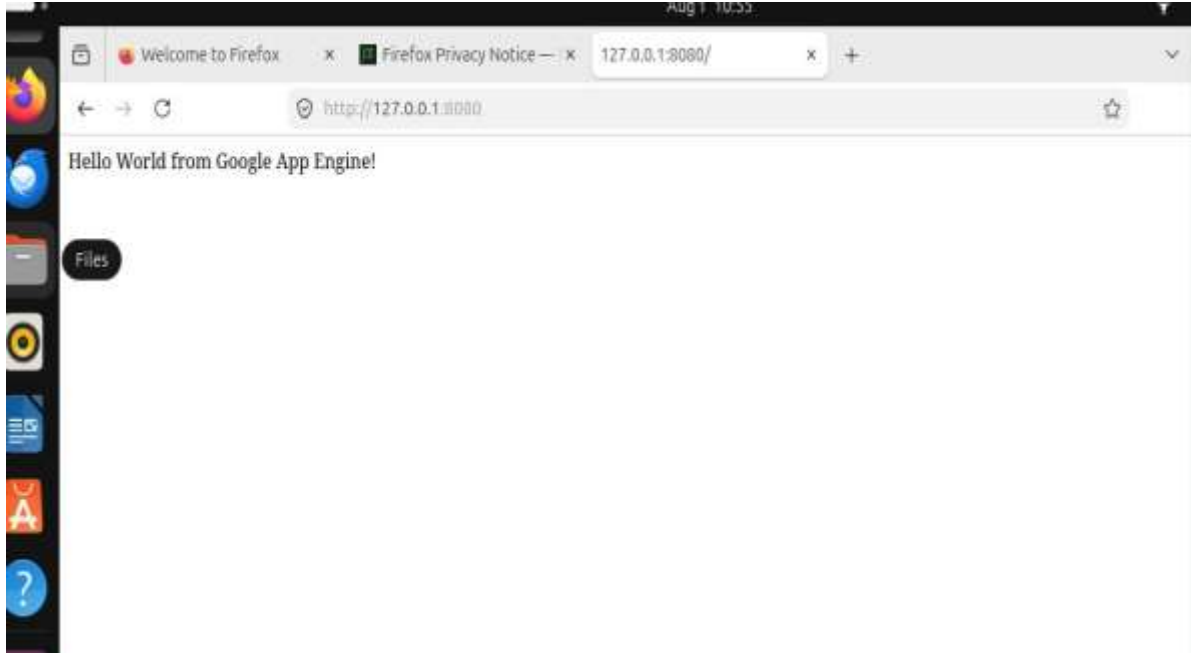
```
(venv) ubuntu@ubuntu:~/gse-hello-world$ pip install --upgrade flask werkzeug
Requirement already satisfied: flask in ./venv/lib/python3.14/site-packages (2.0.1)
Collecting flask
  Downloading flask-3.1.3-py3-none-any.whl.metadata (3.2 kB)
Requirement already satisfied: werkzeug in ./venv/lib/python3.14/site-packages (3.1.0)
Collecting blinker>=1.9.0 (from flask)
  Downloading blinker-1.9.0-py3-none-any.whl.metadata (1.0 kB)
Requirement already satisfied: click>=8.1.3 in ./venv/lib/python3.14/site-packages (from flask) (8.4.2)
Requirement already satisfied: itsdangerous>=2.2.0 in ./venv/lib/python3.14/site-packages (from flask) (2.2.0)
Requirement already satisfied: Jinja2>=3.1.2 in ./venv/lib/python3.14/site-packages (from flask) (3.1.6)
Requirement already satisfied: MarkupSafe>=2.1.1 in ./venv/lib/python3.14/site-packages (from flask) (3.0.3)
Downloading flask-3.1.3-py3-none-any.whl (103 kB)
Downloading blinker-1.9.0-py3-none-any.whl (4.5 kB)
Installing collected packages: blinker, flask
Attempting uninstall: flask
  Found existing installation: flask 2.0.1
  Uninstalling flask-2.0.1:
    Successfully uninstalled flask-2.0.1
Successfully installed blinker-1.9.0 flask-3.1.3
(venv) ubuntu@ubuntu:~/gse-hello-world$ python main.py
* Serving Flask app "main"
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:8080
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 903-333-681
127.0.0.1 - - [01/Aug/2026 10:32:13] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [01/Aug/2026 10:32:13] "GET /favicon.ico HTTP/1.1" 404 -
```

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Harish Kumar V

Output:

The application was accessed from the browser at 127.0.0.1:8080, confirming successful deployment:



Result:

Thus, the Flask-based web application was successfully launched and tested locally using Google App Engine tools.